

Tab 1

Asian University For Women

ICP Final Project Report

Project Title: Campus Compass

Student 1 Details -

Name: Mumtahina Enan

ID: 251011

AUW Email: auw251011@auw.edu.bd

Student 2 Details -

Name: Fatima Rahim Saba

ID: 251005

AUW Email: auw251005@auw.edu.bd

Student 3 Details -

Name: Ishfaqunnesa Tahia

ID: 251007

AUW Email: auw251007@auw.edu.bd

Tab 2

1. Abstract :

Campus Compass is a simple interactive student website designed to help university students manage different aspects of their campus life. The website combines academic tools, well-being features, personal reflection, and a digital time capsule in one platform. It includes an attendance calculator, GPA calculator, mood selection, stress self-check, digital time capsule, and a semester story section.

The project was developed using **HTML, CSS, and JavaScript**. HTML was used to structure the webpage, CSS was used to design and style the interface, and JavaScript was used to add interactive functions and calculations. The final website provides a simple and student-friendly interface that allows users to interact with different features directly from the webpage.

2.Introduction:

University life involves academic responsibilities as well as personal and emotional challenges. Students need to keep track of their attendance and grades while also managing stress and maintaining their well-being. The purpose of Campus Compass is to create a small website that brings some of these useful functions together.

The main objective of the project was to develop an interactive and easy to use website for students. The website contains separate sections for academic tools, well-being, personal reflection, and semester experiences. The HTML file organizes these sections, including the Academic Tools, Well-being, Digital Time Capsule, and Semester Story sections.

Another objective was to make the website visually attractive while keeping the design simple. A university-themed image was added to the header as a background, and different colors, rounded boxes, buttons, and spacing were used to create a clear layout.

Tab 3

3. Methodology / Implementation :

HTML Implementation

HTML was used to create the structure and content of the website. The webpage begins with the title Campus Compass and a header containing the heading "Campus Compass" and the description "Your guide to university life." The website was divided into several main sections. The Academic Tools section contains an Attendance Calculator and GPA Calculator. The Attendance Calculator accepts the number of classes attended and the total number of classes. The GPA Calculator accepts four course grades. The Well-being section allows users to select how they are feeling and also includes a stress self-check. Users can select options such as Happy, Okay, Sad, or Stressed. They can also rate different aspects of their stress from 1 to 5. A Digital Time Capsule was also included. Users can write a message for their future selves, choose a date, and press the button to seal the time capsule.

CSS Implementation

An external CSS file was used to style the webpage. The body was given a light blue background, dark text, and an Arial font.

The header contains a background image, a fixed height, centered text, and padding. This creates a visually attractive introduction to the website.

The main content is organized using boxes with a light orange background, rounded corners, padding, and shadows. This separates the different features and makes the webpage easier to read. Buttons were also styled using CSS. They have a blue background, white text, rounded corners, and a hover effect that changes their background color when the user moves the cursor over them.

JavaScript Implementation

JavaScript was used to make the website interactive.

The attendance function takes the number of attended classes and total classes and calculates the attendance percentage using the formula:

$$\text{Attendance Percentage} = (\text{Classes Attended} \div \text{Total Classes}) \times 100$$

The result is displayed with two decimal places.

The GPA function collects four course grades and calculates their average. The result is then displayed as the user's GPA with two decimal places.

The mood function displays a different message depending on which mood button the user selects.

The stress function adds the three stress ratings and categorizes the result into low, moderate, or higher stress. Different messages are displayed for each category.

Finally, the capsule function checks whether the user has entered both a message and a date. If either is missing, the website asks the user to complete the required fields. Otherwise, it displays a message confirming that the time capsule has been sealed.

Tab 4

4. Results / Output:

The completed website successfully combines several student oriented functions into one webpage. The main output includes:

- A university-themed **Campus Compass header**
- An **Attendance Calculator**
- A **GPA Calculator**
- A **Mood Selection** feature
- A **Stress Self-Check**
- A **Digital Time Capsule**
- A **Semester Story** section
- A styled **footer**

The HTML connects the external CSS file and JavaScript file, allowing the webpage to use both the visual design and interactive functions.

5. Challenges and Limitations:

One challenge during the development of the project was connecting the HTML elements with the JavaScript functions correctly. Each input field needed the correct ID so that JavaScript could retrieve its value and display the appropriate result. Another challenge was designing the webpage so that different sections were visually separated while maintaining a simple layout. CSS properties such as padding, margins, border radius, and box shadows were used to organize the content.

The project also has some limitations. The Digital Time Capsule currently displays a confirmation message but does not actually store the message for future retrieval. The GPA calculator also uses four course grades only, so it does not currently support a variable number of courses. In addition, the stress self-check is a simple calculation and should not be considered a professional assessment.

6. Conclusion and Discussion:

In conclusion, **Campus Compass** successfully demonstrates how HTML, CSS, and JavaScript can be combined to create an interactive student website. The project provides several useful features within a single webpage, including academic calculations, mood selection, stress checking, and personal reflection.

The project also demonstrates important web development concepts such as HTML structure, external CSS styling, JavaScript functions, user input, calculations, conditional statements, and dynamic output.

Overall, the website achieved its main objective of creating a simple and interactive digital guide for university students.

Tab 5

7. Future Work:

There are several ways the project could be improved in the future.

First, the Digital Time Capsule could be connected to a database or browser storage so that messages can actually be saved and accessed on the selected future date.

Second, the GPA Calculator could be expanded to support more courses and different credit hours. It could also calculate a more detailed semester GPA.

The well-being section could also be improved by adding more activities such as breathing exercises, study-break reminders, motivational messages, or links to university support services.

Finally, the website could be made responsive so that it works better on mobile phones and tablets. More animations and interactive elements could also be added to make the user experience more engaging.

8. References:

1. MDN Web Docs. *HTML: HyperText Markup Language*. Mozilla Developer Network.

<https://developer.mozilla.org/en-US/docs/Web/HTML>

2. W3Schools. *JavaScript Tutorial*. <https://www.w3schools.com/js/>

9. Individual Contributions:

Student 1- Was responsible for developing the **HTML structure** of the Campus Compass website. This included creating and organizing the main sections of the webpage, such as the introduction, Academic Tools, Well-being, Digital Time Capsule, Semester Story, and footer. They also created the input fields, buttons, headings, and other HTML elements required for the website's features.

Student 2- Was responsible for the **CSS and visual design** of the website. This included designing the header, adding the background image, choosing colors, styling the content boxes, input fields, buttons, result areas, and footer. They also worked on spacing, sizing, alignment, and the overall appearance of the webpage to make it clear and visually appealing.

Student 3- Was responsible for the **JavaScript and interactive functionality** of the website. This included developing the Attendance Calculator, GPA Calculator, Mood Selection, Stress Self-Check, and Digital Time Capsule functions. They created the calculations, conditions, input handling, and output messages that allow the website to respond to user actions.